

SH-B01 · Prerregistro confirmatorio

Título

SH-B01 · Empirical Contrast of the Structural Hypothesis: Does Explicit Representation of the Material Base Reduce False Positives of Continuity?

Tipo de estudio

Systematic comparative review / retrospective paired diagnostic comparison.

Estado

Confirmatory protocol derived from Structural Hypothesis v1.1 and SH-B01 Empirical Contrast Protocol v0.3.

Research question

In human–technical configurations materially dependent on a sustaining base, does explicitly representing the state and constraints of that material base reduce false-positive assessments of continuity compared with evaluations based only on the internal functioning of the configuration?

Primary hypothesis

$$\boxed{FPR(M_2) < FPR(M_1)}$$

where:

M_1 = evaluation based on the observable functioning or performance of positions 1 · 2 · 3.

M_2 = the same evaluation plus explicit representation of the state, constraints or validated indicators of Θ , the material base required to sustain the target function.

Null hypothesis:

$$FPR(M_2) \geq FPR(M_1)$$

Conceptual origin

The study derives from the provisional structural notation:

$$0+1+2=3 \rightarrow 0'$$

The notation is conceptual rather than arithmetic or causal. Θ denotes the material/territorial base; 1 human agency; 2 technical apparatus, including AI when applicable; 3 relational emergence; and Θ' the subsequent state of the material base after operation and return.

This study does **not** test whether the notation constitutes a universal law. It tests one narrower empirical proposition: whether explicitly representing the sustaining material base improves the detection of non-continuity.

Primary outcome

False Positive Rate of continuity.

A false positive occurs when:

a case is classified as **CONTINUE** at t_0 , but the target function is subsequently shown at t_1 to be materially non-continuable within the predefined horizon because a necessary material condition crosses, approaches or is demonstrated to lie beyond a relevant viability constraint.

False Positive Rate definition

For each diagnostic model:

$$\text{FPR}(\mathbf{M}) = \text{FP}(\mathbf{M}) / [\text{FP}(\mathbf{M}) + \text{TN}(\mathbf{M})]$$

where the denominator contains all cases empirically classified at t_1 as materially non-continuable within the predefined horizon.

FP = the model classified the case as **CONTINUE** at t_0 , but the target function is subsequently shown at t_1 to be materially non-continuable within the predefined horizon because a necessary material condition crosses, approaches, or is demonstrated to lie beyond a relevant viability constraint.

TN = the model classified the case as **CAUTION / ADJUST** or **HALT / NON-VIABLE** at t_0 , and the target function is subsequently shown at t_1 to be materially non-continuable within the predefined horizon.

Primary comparison:

$$\Delta \text{FPR} = \text{FPR}(\mathbf{M}_1) - \text{FPR}(\mathbf{M}_2)$$

Positive values favour H1.

Target function and material dependency

Before Θ is identified, every case must define:

Target function X requires material condition Y during time horizon Z.

X, Y and Z are frozen before examining the outcome at t_1 .

The definition of Θ may not be broadened retrospectively to accommodate the observed outcome.

Eligibility

A case is eligible only when:

- a specific target function can be defined;
- a necessary material dependency can be justified independently of the later outcome;
- a meaningful t_0 and t_1 can be identified;
- M_1 and M_2 can both be reconstructed using information that existed or could reasonably have been available at t_0 ;
- the subsequent material outcome is empirically documented;
- no hypothetical M_1 needs to be invented retrospectively.

Cases are excluded if the real evaluation at t_0 already fully incorporated the same explicit material-base assessment required by M_2 and no separable M_1 can be reconstructed.

Confirmatory domains

Five domains are frozen before case search:

forest management · soil-dependent agriculture · hydropower/reservoir systems · aquaculture · coastal protection

Target:

$\lfloor 8 \text{ cases/domain} = 40 \text{ cases} \rfloor$

If fewer than eight eligible cases exist in a domain after the predefined search is exhausted, the deficit will be reported and **will not be replaced by another domain**.

Prior cases excluded

The following cases or case families contributed to conceptual development or pilot testing and are excluded from the confirmatory sample:

irrigation-efficiency/rebound cases; fisheries used during development, including Northern cod; data-centre/energy cases; Aral Sea; Montreal Protocol/ozone; and flood-protection cases examined during conceptual development or pilot testing.

They may be referenced as developmental material but cannot contribute to the confirmatory result.

Predefined search strategy

The confirmatory literature search is frozen before any confirmatory case is formally analysed.

Information sources

Two primary scholarly discovery sources will be searched:

1. OpenAlex Works.
2. Semantic Scholar Academic Graph.

Crossref metadata may be used only to verify bibliographic metadata and DOI identifiers; it will not constitute an independent case-discovery source.

Searches will be executed after submission of this preregistration.

Publication date: no lower publication-date limit will be applied. Only evidence published on or before **8 October 2026** will be eligible for the confirmatory search.

Eligible publication languages: English and Spanish.

The same conceptual search blocks will be used across both primary discovery sources. Platform-specific syntax may be adapted only where technically necessary; no search concept may be added or removed after preregistration without a documented amendment.

Domain 1 — Forest management

("forest management" OR forestry OR silviculture OR logging OR "forest restoration" OR "gestión forestal" OR "aprovechamiento forestal" OR "restauración forestal") AND (yield OR harvest OR productivity OR production OR performance OR rendimiento OR cosecha OR productividad OR producción) AND (soil OR water OR biomass OR regeneration OR mortality OR "forest condition" OR suelo OR agua OR biomasa OR regeneración OR mortalidad OR "estado forestal") AND (monitoring OR longitudinal OR "time series" OR "case study" OR trend OR seguimiento OR "serie temporal" OR "estudio de caso" OR tendencia)

Domain 2 — Soil-dependent agriculture

(agriculture OR farming OR cropping OR "crop production" OR agricultura OR cultivo OR "producción agrícola") AND (yield OR productivity OR production OR efficiency OR rendimiento OR productividad OR producción OR eficiencia) AND ("soil organic carbon" OR "soil organic matter" OR erosion OR salinity OR fertility OR "soil moisture" OR nutrients OR "carbono orgánico del suelo" OR "materia orgánica del suelo" OR erosión OR salinidad OR fertilidad OR "humedad del suelo" OR nutrientes) AND (monitoring OR longitudinal OR "time series" OR "case study" OR trend OR seguimiento OR "serie temporal" OR "estudio de caso" OR tendencia)

Domain 3 — Hydropower / reservoir systems

(hydropower OR hydroelectric OR reservoir OR "hydroelectric power" OR hidroelectricidad OR hidroeléctrica OR embalse OR reservorio) AND (generation OR output OR capacity OR reliability OR performance OR generación OR producción OR capacidad OR fiabilidad OR rendimiento) AND (inflow OR storage

OR sedimentation OR "water availability" OR drought OR "reservoir level" OR caudal OR almacenamiento OR sedimentación OR "disponibilidad de agua" OR sequía OR "nivel del embalse") AND (monitoring OR longitudinal OR "time series" OR "case study" OR trend OR seguimiento OR "serie temporal" OR "estudio de caso" OR tendencia)

Domain 4 — Aquaculture

(aquaculture OR "fish farming" OR "shrimp farming" OR mariculture OR acuicultura OR piscicultura OR camaronicultura OR maricultura) AND (yield OR production OR productivity OR efficiency OR performance OR rendimiento OR producción OR productividad OR eficiencia) AND ("water quality" OR oxygen OR nutrients OR feed OR disease OR "carrying capacity" OR temperature OR "calidad del agua" OR oxígeno OR nutrientes OR alimento OR enfermedad OR "capacidad de carga" OR temperatura) AND (monitoring OR longitudinal OR "time series" OR "case study" OR trend OR seguimiento OR "serie temporal" OR "estudio de caso" OR tendencia)

Domain 5 — Coastal protection

("coastal protection" OR "coastal defence" OR "coastal defense" OR seawall OR breakwater OR "beach nourishment" OR "flood defence" OR "flood defense" OR "protección costera" OR "defensa costera" OR "dique costero" OR rompeolas OR "regeneración de playas" OR "defensa contra inundaciones") AND (performance OR effectiveness OR protection OR exposure OR development OR rendimiento OR eficacia OR protección OR exposición OR desarrollo) AND (erosion OR shoreline OR sediment OR subsidence OR "sea level" OR overtopping OR habitat OR erosión OR "línea de costa" OR sedimento OR subsidencia OR "nivel del mar" OR sobrepaso OR hábitat) AND (monitoring OR longitudinal OR "time series" OR "case study" OR trend OR seguimiento OR "serie temporal" OR "estudio de caso" OR tendencia)

Screening and deduplication

All records returned by the preregistered searches will be retained in the search log. Records will first be deduplicated by DOI. Records without a DOI will be deduplicated using normalized title, publication year, and first author.

Title/abstract screening will determine whether a record concerns the relevant confirmatory domain and contains empirical evidence potentially permitting reconstruction of t_0 and t_1 .

Potentially eligible records will undergo full-text screening using the six preregistered eligibility criteria.

The unit of analysis is the empirical case, not the publication. Multiple publications describing the same empirical case will be consolidated into one case record while all contributing sources remain traceable.

Evidence chronology will remain separated into categories A, B, and C. Category C evidence cannot be used to construct M_1 or M_2 .

Citation searching

For each case passing full-text eligibility screening, one generation of backward-reference searching and forward-citation searching will be performed using OpenAlex and/or Semantic Scholar. Records identified through citation searching must satisfy the same eligibility and chronology rules.

Search exhaustion

A domain search will be considered exhausted only when:

- (a) all records returned by both preregistered discovery sources have been screened;
- (b) duplicates have been resolved;
- (c) full texts of potentially eligible records have been assessed where obtainable; and
- (d) one generation of backward and forward citation searching for eligible cases has been completed.

Case selection when more than eight eligible cases exist

If more than eight eligible empirical cases remain in a confirmatory domain after search exhaustion and case consolidation, eight cases will be selected by deterministic random sampling without replacement.

Eligible cases will first receive stable identifiers based on DOI; where no DOI exists, normalized title plus publication year will be used.

The fixed random seed will be **20261008**.

All eligible cases not selected for the confirmatory eight-case sample will remain visible in the search and eligibility log and may be reported only as secondary/exploratory material. They will not be selectively substituted into the confirmatory sample.

Search documentation

For every search, the following will be retained: source, exact query, execution date, number of records retrieved, deduplication results, screening decisions, exclusion reasons, citation-search additions, and final eligible-case count.

Any modification after preregistration will require a dated protocol amendment stating the exact change, reason for the change, and whether any confirmatory evidence had already been inspected.

Information chronology

Three categories must be recorded separately:

A. Available at t_0 — information genuinely available before the outcome.

B. Retrospective reconstruction — later sources reproducing measurements or records that already existed at t_0 .

C. Later interpretation — causal or diagnostic understanding developed only after t_1 .

Category C cannot be used to construct M_1 or M_2 .

Coding sequence

For every eligible case:

Select case

↓

Freeze target function X

↓

Freeze material dependency Y

↓

Freeze horizon Z

↓

Open t_0 evidence only

↓

Code M_1

↓

Code M_2

↓

Lock both classifications

↓

Open t_1 evidence

↓

Classify observed outcome

↓

Score FP / TN / etc.

Diagnostic categories

At t_0 :

CONTINUE — sufficient evidence supports material continuity within the defined conditions and horizon.

CAUTION / ADJUST — uncertainty or deterioration prevents an unconditional continuity judgment.

HALT / NON-VIABLE — evidence indicates incompatibility with continued operation under the defined conditions.

For the primary FPR analysis, only **CONTINUE** counts as a positive continuity prediction.

Primary analysis

Because M_1 and M_2 evaluate the same cases, the comparison is paired.

Primary statistics:

$[FPR(M_1), \text{quad } FPR(M_2), \text{quad } \Delta FPR]$

A two-sided exact McNemar test will be applied to the discordant paired classifications and reported as secondary inferential information. Raw discordant counts will always be reported. If there are no discordant pairs, this absence of discordance will be reported explicitly. The McNemar test is not the primary decision criterion and will not replace the preregistered FPR comparison.

Interpretation rules

PASS provisional if $FPR(M_2) < FPR(M_1)$ across the confirmatory sample without retrospective redefinition of function, material base or horizon.

NO PASS if $FPR(M_2) \geq FPR(M_1)$, or if any apparent advantage of M_2 depends systematically on retrospective information or flexible reclassification.

UNRESOLVED if the comparison cannot be interpreted because of insufficient data, ambiguous dependencies, temporal incompatibility or excessive unclassifiable cases.

All UNRESOLVED cases remain visible in the complete dataset.

Secondary analyses

Secondary analyses may examine false negatives, proportion of CAUTION classifications, domain heterogeneity and sensitivity to alternative legitimate thresholds.

They will be labelled **secondary/exploratory** and cannot replace the primary test if the primary result is unfavourable.

Independent coding

A second independent coder should evaluate a preregistered random **25% subset** of confirmatory cases.

Agreement should be reported for: target function · material dependency · M_1 · M_2 · t_1 outcome.

If independent coding is not obtained, results must be described as **internal confirmatory evidence**, not independent scientific validation.

Amendment rule

Once this preregistration is submitted:

No confirmatory rule will be silently changed.

Any modification must receive:

- amendment identifier;
- date;
- reason;
- exact changed text;
- indication of whether any confirmatory case had already been inspected;
- impact on confirmatory versus exploratory status.

Stopping rule

The study ends after all predefined searches have been completed and up to **40 eligible cases** have been coded.

The study will **not** stop early because a favourable or unfavourable statistical result has appeared.

Data transparency

The final report will publish:

- complete included-case table;
- exclusions and reasons;
- M_1 and M_2 classifications;
- relevant t_0 evidence;
- t_1 outcome;
- UNRESOLVED cases;
- protocol amendments;
- analysis code or calculations where applicable.

Negative and contradictory cases will not be removed.

Prior knowledge declaration

This preregistration follows substantial conceptual development and exploratory testing of the Structural Hypothesis. Several known cases were examined while developing and piloting SH-B01. Those cases are explicitly excluded from the confirmatory sample. No case from the five frozen confirmatory domains will be analysed for the confirmatory test before submission of this preregistration.

Robles Rionegro, D. A. (2026). SH-B01 · Empirical Contrast of the Structural Hypothesis: Does Explicit Representation of the Material Base Reduce False Positives of Continuity? (Version 0.3). OSF. <https://doi.org/10.17605/OSF.IO/BCT9S>

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